

ON THE ROLE OF SOURCE DENSITY IN HYDRA PATTERNING AND REGENERATION MODELS

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ABSTRACT. • Problem: source density is often used in Hydra models, but its precise mathematical and biological role is not clear.

- Different Gierer–Meinhardt model variants have different properties, even when they are treated as versions of the same model.
- The goal of the paper is to separate what follows from the Gierer–Meinhardt mechanism itself from what genuinely requires source density.
- We show that source density is not necessary for basic regeneration or classical pattern formation.
- Source density becomes important when modelling positional memory and grafting/polarization experiments.
- Dynamic source density can generate long-lived transients or metastable peaks, but it does not necessarily produce stable stationary states at arbitrary positions.
- We propose an extension with a hysteretic head identity variable.

1. INTRODUCTION

- Hydra as a classical model organism for regeneration and pattern formation.
- Head regeneration and the emergence of localized organizer/head signals.
- Activator-inhibitor modelling tradition in Hydra.
- Gierer–Meinhardt models and their role in explaining head formation.
- Ambiguity: several different mathematical systems are called Gierer–Meinhardt-type models.
- Importance of distinguishing:
 - models with constitutive production;
 - models without constitutive production;
 - models with source-dependent production;
 - models with prescribed source density;
 - models with dynamic source density.
- Biological problem: regeneration requires wound response. Therefore, a reasonable model should allow a stable resting/no-head state and activation only after sufficiently strong perturbation.
- Pattern formation problem: classical diffusion-driven instability and Turing mechanisms can generate peaks, but often select peak locations by domain geometry and unstable modes.
- Positional memory problem: grafting and polarization experiments suggest that tissue fragments remember where they came from.
- Source density/source gradient as a candidate mechanism for positional information.
- Main questions of the paper:
 - What does source density add to Gierer–Meinhardt-type Hydra models?
 - Is it needed for regeneration?

- Is it needed for positional memory?
- Is it enough to stabilize heads at arbitrary graft-induced locations?
- Outline of contributions:
 - systematic audit of Gierer–Meinhardt variants;
 - comparison of stability and diffusion-driven instability properties;
 - comparison of regeneration and pattern formation with and without source density;
 - analysis of source density as positional memory;
 - proposal of a head identity variable with hysteresis.

2. REVIEW OF GIERER–MEINHARDT-TYPE MODELS

This section records the model variants that will be compared in the paper. The point is not to rederive the general theory of diffusion-driven instability, but to identify which qualitative properties are already present in the underlying Gierer–Meinhardt equations and which properties are introduced by source density.

2.1. Two-Equation Activator–Inhibitor Models. Throughout this section, we assume that $a, b, \mu_u, \mu_v, D_u, D_v > 0$, while the production parameters p_u, p_v, q_u are nonnegative.

We start from the broad two-equation Gierer–Meinhardt activator-inhibitor family

$$(2.1) \quad \begin{aligned} \frac{\partial u}{\partial t} &= D_u \Delta u + \frac{au^2 + q_u}{v} - \mu_u u + p_u, \\ \frac{\partial v}{\partial t} &= D_v \Delta v + bu^2 - \mu_v v + p_v. \end{aligned}$$

Here u denotes the activator, v denotes the inhibitor, and homogeneous Neumann boundary conditions are imposed. The terms p_v , p_u , and q_u play different formal roles:

- p_v is basal inhibitor production;
- p_u is additive activator production;
- q_u is activator production inside the inhibitor-regulated term.

The qualitative behaviour of the model depends strongly on which of these terms are present. Basal inhibitor production as a denominator shift. The term p_v is equivalent to adding a positive constant to the inhibitor denominator. Indeed, writing the original inhibitor variable as V and setting

$$(2.2) \quad V = v + K \quad \text{with} \quad K := \frac{p_v}{\mu_v},$$

the system can be written equivalently as

$$(2.3) \quad \begin{aligned} \frac{\partial u}{\partial t} &= D_u \Delta u + \frac{au^2 + q_u}{K + v} - \mu_u u + p_u, \\ \frac{\partial v}{\partial t} &= D_v \Delta v + bu^2 - \mu_v v. \end{aligned}$$

In this shifted variable, $v = 0$ corresponds to the basal inhibitor level $V = K$.

2.1.1. No production terms: $p_v = p_u = q_u = 0$. If all production terms are absent, the model is

$$(2.4) \quad \begin{aligned} \partial_t u &= D_u \Delta u + \frac{au^2}{v} - \mu_u u, \\ \partial_t v &= D_v \Delta v + bu^2 - \mu_v v. \end{aligned}$$

Proposition 2.1. *The system (2.4) has exactly one positive spatially homogeneous steady state, given by*

$$(2.5) \quad \bar{u} = \frac{a\mu_v}{b\mu_u}, \quad \bar{v} = \frac{a^2\mu_v}{b\mu_u^2}.$$

This steady state is unstable if $\mu_u > \mu_v$. If $\mu_u < \mu_v$, it may be stable or exhibit a Turing instability if D_u is sufficiently small relative to D_v .

Proof. Let $(\bar{u}, \bar{v})$ be a spatially homogeneous steady state. The diffusion terms vanish, and the second equation in (2.4) gives

$$(2.6) \quad \bar{v} = \frac{b}{\mu_v} \bar{u}^2.$$

If $\bar{u} = 0$, then $\bar{v} = 0$, but $(0, 0)$ is not a steady state because the term $a\bar{u}^2/\bar{v}$ is undefined there. Hence $\bar{u} \neq 0$ and $\bar{v} > 0$. The first steady-state equation can therefore be divided by $\bar{u}$, yielding

$$(2.7) \quad \frac{a\bar{u}}{\bar{v}} = \mu_u.$$

Since $\bar{v} > 0$, this identity also implies that $\bar{u} > 0$. Substitution of $\bar{v} = b\bar{u}^2/\mu_v$ gives

$$(2.8) \quad \bar{u} = \frac{a\mu_v}{b\mu_u}.$$

The corresponding value of $\bar{v}$ is

$$\bar{v} = \frac{b}{\mu_v} \left(\frac{a\mu_v}{b\mu_u} \right)^2 = \frac{a^2\mu_v}{b\mu_u^2}.$$

The Jacobian at this steady state is

$$(2.9) \quad J(\bar{u}, \bar{v}) = \begin{pmatrix} \mu_u & -\frac{\mu_u^2}{a} \\ \frac{2a\mu_v}{\mu_u} & -\mu_v \end{pmatrix},$$

and hence

$$(2.10) \quad \text{tr } J(\bar{u}, \bar{v}) = \mu_u - \mu_v, \quad \det J(\bar{u}, \bar{v}) = \mu_u\mu_v > 0.$$

Since $\det J(\bar{u}, \bar{v}) > 0$, the sign of $\text{tr } J(\bar{u}, \bar{v})$ gives the stated stability properties.

Moreover, the activator self-interaction satisfies

$$\partial_u f(\bar{u}, \bar{v}) = \mu_u > 0.$$

Therefore, by the criterion in [XXX], a Turing instability may occur when D_u is sufficiently small relative to D_v . $\square$

Remark 2.2. Model (2.4) has no resting state and therefore no bistability between resting and pattern-forming states. For fixed parameters, it either generates spatial patterns or does not generate patterns at all.

2.1.2. Inhibitor production only: $p_v > 0$, $p_u = q_u = 0$. Adding basal inhibitor production removes the singularity. By the change of variables above, the system takes the saturated form

$$(2.11) \quad \begin{aligned} \frac{\partial u}{\partial t} &= D_u \Delta u + \frac{au^2}{K+v} - \mu_u u, \\ \frac{\partial v}{\partial t} &= D_v \Delta v + bu^2 - \mu_v v. \end{aligned}$$

This will be the reference two-equation model in the later subsections. After rescaling one may set $K = 1$, but we keep K visible because it records the effect of inhibitor production.

Proposition 2.3. *Under the condition $a^2 > 4b\mu_u^2 K/\mu_v$, the system (2.11) has three spatially homogeneous steady states with the following stability properties:*

$$\begin{aligned} (0, 0) & \quad \text{asymptotically stable,} \\ (\bar{u}_-, \bar{v}_-) & \quad \text{unstable,} \\ (\bar{u}_+, \bar{v}_+) & \quad \begin{aligned} & \text{if } \mu_u > \mu_v, \text{ then unstable,} \\ & \text{if } \mu_u < \mu_v, \text{ then either stable or Turing unstable.} \end{aligned} \end{aligned}$$

If $a^2 < 4b\mu_u^2 K/\mu_v$, then $(0, 0)$ is the unique spatially homogeneous steady state and is asymptotically stable.

Proof. At the zero state, the Jacobian is

$$(2.12) \quad J(0, 0) = \begin{pmatrix} -\mu_u & 0 \\ 0 & -\mu_v \end{pmatrix}.$$

Its eigenvalues are negative, and diffusion shifts them further to the left. Thus the zero state is asymptotically stable.

Let $(\bar{u}, \bar{v})$ be a positive spatially homogeneous steady state. The steady-state equations give

$$(2.13) \quad \bar{v} = \frac{b}{\mu_v} \bar{u}^2, \quad \frac{a\bar{u}}{K + \bar{v}} = \mu_u.$$

Substitution of $\bar{v} = b\bar{u}^2/\mu_v$ yields

$$(2.14) \quad \frac{\mu_u b}{\mu_v} \bar{u}^2 - a\bar{u} + \mu_u K = 0.$$

This quadratic equation has two distinct positive roots precisely when its discriminant is positive, that is, when $a^2 > 4b\mu_u^2 K/\mu_v$. Denote these roots by $\bar{u}_- < \bar{u}_+$ and set $\bar{v}_\pm = b\bar{u}_\pm^2/\mu_v$. If the inequality is reversed, the quadratic has no real roots, and hence the zero state is the only spatially homogeneous steady state.

The Jacobian at either positive steady state is

$$(2.15) \quad J(\bar{u}_\pm, \bar{v}_\pm) = \begin{pmatrix} \mu_u & -\frac{\mu_u^2}{a} \\ 2b\bar{u}_\pm & -\mu_v \end{pmatrix}.$$

Its trace and determinant are

$$(2.16) \quad \text{tr } J(\bar{u}_\pm, \bar{v}_\pm) = \mu_u - \mu_v, \quad \det J(\bar{u}_\pm, \bar{v}_\pm) = \mu_u \left(\frac{2b\mu_u}{a} \bar{u}_\pm - \mu_v \right).$$

If P denotes the polynomial on the left-hand side of (2.14), then

$$\det J(\bar{u}_\pm, \bar{v}_\pm) = \frac{\mu_u \mu_v}{a} P'(\bar{u}_\pm).$$

By strict convexity of P , $P'(\bar{u}_-) < 0 < P'(\bar{u}_+)$. Hence the lower state is unstable, while the sign of the trace gives the stated stability properties of the upper state. Moreover,

$$\partial_u f(\bar{u}_+, \bar{v}_+) = \mu_u > 0.$$

Therefore, by the criterion in [XXX], the upper state may exhibit a Turing instability when D_u is sufficiently small relative to D_v . $\square$

Remark 2.4. Model (2.11) has a stable zero state, which may serve as a resting state and makes bistability with the upper positive state possible.

2.1.3. *Activator and inhibitor production:* $p_v > 0$, $p_u, q_u \geq 0$. With inhibitor production already encoded by $K > 0$, the two-equation model with activator production is

$$(2.17) \quad \begin{aligned} \frac{\partial u}{\partial t} &= D_u \Delta u + \frac{au^2 + q_u}{K + v} - \mu_u u + p_u, \\ \frac{\partial v}{\partial t} &= D_v \Delta v + bu^2 - \mu_v v. \end{aligned}$$

This model interpolates between the two preceding cases. For sufficiently weak activator production, the three-state structure may persist. As production increases, two states may vanish, leaving a single positive spatially homogeneous steady state.

Set

$$(2.18) \quad A := \frac{\mu_u b}{\mu_v}, \quad B := a + \frac{p_u b}{\mu_v}, \quad C := \mu_u K, \quad D := q_u + K p_u,$$

and let

$$(2.19) \quad \Delta := B^2 C^2 - 4AC^3 - 4B^3 D - 27A^2 D^2 + 18ABCD$$

be the discriminant of $P(z) = Az^3 - Bz^2 + Cz - D$.

For any simple spatially homogeneous steady state $(\bar{u}_i, \bar{v}_i)$, define

$$(2.20) \quad S_i := \partial_u f(\bar{u}_i, \bar{v}_i) = \frac{2a\bar{u}_i}{K + \bar{v}_i} - \mu_u.$$

Proposition 2.5. *Under the condition $\Delta > 0$, the system (2.17) has three distinct nonnegative spatially homogeneous steady states, ordered by $\bar{u}_1 < \bar{u}_2 < \bar{u}_3$, with the following properties:*

	<i>if $S_1 \leq 0$, then stable,</i>
$(\bar{u}_1, \bar{v}_1)$	<i>if $0 < S_1 < \mu_v$, then either stable or Turing unstable,</i>
	<i>if $S_1 > \mu_v$, then unstable,</i>
$(\bar{u}_2, \bar{v}_2)$	<i>unstable,</i>
	<i>if $S_3 \leq 0$, then stable,</i>
$(\bar{u}_3, \bar{v}_3)$	<i>if $0 < S_3 < \mu_v$, then either stable or Turing unstable,</i>
	<i>if $S_3 > \mu_v$, then unstable.</i>

Moreover, the inequality $S_1 < S_3$ holds, showing that the upper state is less stable with respect to the reaction dynamics.

If $\Delta < 0$, the system has exactly one nonnegative spatially homogeneous steady state. It is stable if $S_1 \leq 0$, may be stable or Turing unstable if $0 < S_1 < \mu_v$, and is unstable if $S_1 > \mu_v$.

Proof. At a spatially homogeneous steady state,

$$(2.21) \quad \bar{v} = \frac{b}{\mu_v} \bar{u}^2, \quad P(\bar{u}) = A\bar{u}^3 - B\bar{u}^2 + C\bar{u} - D = 0.$$

Since $D \geq 0$, one has $P(0) \leq 0$ and $P(z) \rightarrow \infty$ as $z \rightarrow \infty$. Moreover, $P(z) < 0$ for every $z < 0$. Hence all real roots of P are nonnegative, and the standard discriminant criterion gives the stated alternatives for Δ . If $D > 0$, then all real roots are positive.

The Jacobian at $(\bar{u}_i, \bar{v}_i)$ is

$$(2.22) \quad J(\bar{u}_i, \bar{v}_i) = \begin{pmatrix} \frac{2a\bar{u}_i}{K + \bar{v}_i} - \mu_u & -\frac{a\bar{u}_i^2 + q_u}{(K + \bar{v}_i)^2} \\ 2b\bar{u}_i & -\mu_v \end{pmatrix}.$$

Writing $h(u) = f(u, bu^2/\mu_v)$, one has

$$h(u) = -\frac{P(u)}{K + bu^2/\mu_v}.$$

It follows that

$$(2.23) \quad \text{tr } J(\bar{u}_i, \bar{v}_i) = S_i - \mu_v, \quad \det J(\bar{u}_i, \bar{v}_i) = \frac{\mu_v}{K + \bar{v}_i} P'(\bar{u}_i).$$

For three simple roots,

$$P'(\bar{u}_1) > 0, \quad P'(\bar{u}_2) < 0, \quad P'(\bar{u}_3) > 0.$$

Thus the middle state is unstable, whereas the determinant is positive at both outer states. When $\Delta < 0$, the unique real root is simple and satisfies $P'(\bar{u}_1) > 0$, so the determinant is also positive at the unique steady state.

When $\Delta > 0$, Vieta's relations give

$$\bar{u}_1 \bar{u}_2 + \bar{u}_1 \bar{u}_3 + \bar{u}_2 \bar{u}_3 = \frac{C}{A} = \frac{K\mu_v}{b}.$$

Thus $\bar{u}_1 \bar{u}_3 < K\mu_v/b$, and a direct calculation yields

$$S_3 - S_1 = \frac{2a(\bar{u}_3 - \bar{u}_1) \left(K - \frac{b}{\mu_v} \bar{u}_1 \bar{u}_3 \right)}{\left(K + \frac{b}{\mu_v} \bar{u}_1^2 \right) \left(K + \frac{b}{\mu_v} \bar{u}_3^2 \right)} > 0.$$

Since $\text{tr } J(\bar{u}_i, \bar{v}_i) = S_i - \mu_v$, an outer or unique state is unstable if $S_i > \mu_v$, while its reaction Jacobian is stable if $S_i < \mu_v$.

For a Laplacian eigenvalue $\lambda \geq 0$,

$$(2.24) \quad \begin{aligned} \text{tr}(J(\bar{u}_i, \bar{v}_i) - \lambda \text{diag}(D_u, D_v)) &= S_i - \mu_v - \lambda(D_u + D_v), \\ \det(J(\bar{u}_i, \bar{v}_i) - \lambda \text{diag}(D_u, D_v)) &= \det J(\bar{u}_i, \bar{v}_i) + \lambda(\mu_v D_u - D_v S_i) + \lambda^2 D_u D_v. \end{aligned}$$

For an outer or unique state with $S_i \leq 0$, both expressions have the signs required for stability for every $\lambda \geq 0$. If $0 < S_i < \mu_v$, the reaction Jacobian is stable, while the criterion in [XXX] allows a Turing instability when D_u is sufficiently small relative to D_v . This proves all three stability alternatives.

Finally, at any positive steady state, the steady-state identity gives

$$S_i = \mu_u - \frac{2p_u}{\bar{u}_i} - \frac{2q_u}{\bar{u}_i(K + \bar{v}_i)} \leq \mu_u.$$

Thus $S_i < \mu_v$ whenever $\mu_u < \mu_v$. □

Remark 2.6. For the reaction dynamics, the lower state cannot be unstable while the upper state is stable. The opposite configuration, with a stable lower state and an unstable upper state, is possible.

Remark 2.7. If $p_u > 0$ or $q_u > 0$, activator production removes the zero steady state. In the three-state regime, the lower positive state may instead act as a resting state, allowing bistability with the upper positive state.

2.2. Reference Two-Equation Model for Regeneration. The reference two-equation model for the regeneration discussion is therefore (2.11), the inhibitor-production model without activator production. It has:

- a stable zero/no-head state;
- two positive steady states when $a^2 > 4b\mu_u^2 K/\mu_v$;
- a lower positive branch that is unstable;
- an upper positive branch that is kinetically stable when $\mu_u < \mu_v$ and can become unstable in the presence of diffusion.

This model already contains the basic threshold logic needed for wound-induced regeneration. Source density is therefore not needed for the existence of a stable resting state or a threshold to activation.

2.3. Three-Equation Model with Source Density. We now add a source-density variable s . In the shifted inhibitor variable, the source-density model with both activator production mechanisms is

$$(2.25) \quad \begin{aligned} \frac{\partial u}{\partial t} &= D_u \Delta u + s \frac{au^2 + q_u}{K + v} - \mu_u u + p_u, \\ \frac{\partial v}{\partial t} &= D_v \Delta v + bsu^2 - \mu_v v, \\ \tau \frac{\partial s}{\partial t} &= D_s \Delta s + u - s. \end{aligned}$$

Here $K > 0$ again represents basal inhibitor production. The term q_u is source-dependent because it is multiplied by s , whereas p_u is an external activator production term.

2.3.1. *No activator production:* $p_u = q_u = 0$. When $p_u = q_u = 0$, the model is

$$(2.26) \quad \begin{aligned} \partial_t u &= D_u \Delta u + s \frac{au^2}{K + v} - \mu_u u, \\ \partial_t v &= D_v \Delta v + bsu^2 - \mu_v v, \\ \tau \partial_t s &= D_s \Delta s + u - s. \end{aligned}$$

The zero state $(0, 0, 0)$ is a homogeneous steady state. The reaction Jacobian at zero is triangular up to the u -to- s coupling:

$$(2.27) \quad J(0, 0, 0) = \begin{pmatrix} -\mu_u & 0 & 0 \\ 0 & -\mu_v & 0 \\ 1/\tau & 0 & -1/\tau \end{pmatrix},$$

so the zero state is stable.

For a positive homogeneous steady state, the third equation gives $s = u$, and the second equation gives

$$(2.28) \quad v = \frac{b}{\mu_v} u^3.$$

The first equation then yields

$$(2.29) \quad \frac{\mu_u b}{\mu_v} u^3 - au^2 + \mu_u K = 0.$$

Thus source density changes the positive steady-state equation from the quadratic equation (2.14) to a cubic equation. Two positive steady states exist when the minimum of the cubic is negative, equivalently

$$(2.30) \quad \mu_u K < \frac{4a^3 \mu_v^2}{27 \mu_u^2 b^2}.$$

At a positive steady state, the reaction Jacobian in the variable order (u, v, s) is

$$(2.31) \quad \begin{aligned} J(u, v, s) &= \begin{pmatrix} \mu_u & -\frac{\mu_u^2}{au} & \mu_u \\ 2bu^2 & -\mu_v & bu^2 \\ 1/\tau & 0 & -1/\tau \end{pmatrix}, \\ s &= u, \quad v = \frac{b}{\mu_v} u^3. \end{aligned}$$

The characteristic polynomial has the form

$$(2.32) \quad \chi(\lambda) = \lambda^3 + \alpha_1 \lambda^2 + \alpha_2 \lambda + \alpha_3,$$

where

$$(2.33) \quad \begin{aligned} \alpha_1 &= \mu_v + \frac{1}{\tau} - \mu_u, \\ \alpha_2 &= \frac{\mu_v - 2\mu_u}{\tau} - \mu_u \mu_v + \frac{2b\mu_u^2}{a}u, \\ \alpha_3 &= \frac{\mu_u}{\tau} \left(\frac{3b\mu_u}{a}u - 2\mu_v \right). \end{aligned}$$

The lower positive branch lies below the critical value

$$(2.34) \quad u_c = \frac{2a\mu_v}{3b\mu_u},$$

so $\alpha_3 < 0$ and that branch is unstable. The upper positive branch lies above u_c , so it is the only candidate for kinetic stability. It is stable for the homogeneous kinetics precisely when the Routh–Hurwitz inequalities

$$(2.35) \quad \begin{aligned} \alpha_1 &> 0, \\ \alpha_2 &> 0, \\ \alpha_3 &> 0, \\ \alpha_1 \alpha_2 &> \alpha_3 \end{aligned}$$

hold. With suitable time scale and degradation parameters, this upper branch is the source-density analogue of the upper pattern-forming branch of the two-equation reference model (2.11).

2.3.2. Source-dependent activator production: $q_u > 0$, $p_u = 0$. If $q_u > 0$ but $p_u = 0$, the zero state remains a steady state because the extra activator production is multiplied by s . Its u, s linearization is

$$(2.36) \quad \begin{pmatrix} -\mu_u & q_u/K \\ 1/\tau & -1/\tau \end{pmatrix}.$$

Thus the zero state remains stable when

$$(2.37) \quad q_u < \mu_u K,$$

and loses stability when q_u is too large.

For positive homogeneous states, $s = u$, $v = bu^3/\mu_v$, and

$$(2.38) \quad \frac{\mu_u b}{\mu_v} u^3 - au^2 + \mu_u K - q_u = 0.$$

Small q_u lowers the activation threshold but can preserve the resting state. Large q_u destroys the threshold-regeneration interpretation because the zero state is no longer stable.

2.3.3. External activator production: $p_u > 0$. If $p_u > 0$, then $(0, 0, 0)$ is no longer a steady state. Positive homogeneous states satisfy $s = u$, $v = bu^3/\mu_v$, and

$$(2.39) \quad 0 = u(au^2 + q_u) + (p_u - \mu_u u) \left(K + \frac{b}{\mu_v} u^3 \right).$$

This is generally a quartic equation in u . For small p_u , the behaviour is expected to remain close to the threshold case: the zero state is replaced by a small positive resting state, and large perturbations may reach the upper pattern-forming branch. For sufficiently large p_u , the lower resting branch and the threshold branch may disappear, leaving only a positive pattern-forming state. In that regime the model produces patterns without requiring a wound-like trigger.

TABLE 1. Qualitative audit of the model variants used in this paper.

Model	Resting state	Positive branches	Unstable branch	Interpretation
Two equations, no production	No regular zero	One positive branch	None	Pattern-forming model without a no-head state; can produce large spikes.
Two equations, inhibitor production only	Stable zero	Two positive branches	Lower positive branch	Threshold-regeneration mechanism; upper branch is the pattern-forming candidate.
Two equations, small activator production	Small positive rest state	Typically three positive states	Middle branch	Threshold behaviour persists approximately.
Two equations, large activator production	No resting/threshold regime	Typically one positive state	None	Patterning may occur without a wound-like perturbation.
Source density, no activator production	Stable zero	Two positive branches when (2.30) holds	Lower positive branch	Similar threshold logic, but the steady-state equation and stability conditions change.
Source density, source-dependent q_u	Stable zero if $q_u < \mu_u K$	Production-dependent	Production-dependent	Small q_u lowers the threshold; large q_u can destabilize the resting state.
Source density, external p_u	Zero absent	Production-dependent	Production-dependent	Small p_u gives a small positive rest state; large p_u removes the threshold regime.

2.4. Summary of the Audit. The conclusion of this audit is that the inhibitor-production version of the two-equation Gierer–Meinhardt model already gives a stable resting state, a threshold to activation, and a pattern-forming upper branch. Source density does not need to be introduced to obtain basic regeneration-like threshold dynamics. Its role should therefore be tested in the later sections as a mechanism for positional memory, grafting response, and long-lived anchoring of head-forming regions.

3. REGENERATION AND PATTERN FORMATION

We now ask what can already be explained by the two-equation reference model (2.11), before introducing source density. This is the minimal test that the model should pass: if basic regeneration and peak formation already follow from the activator-inhibitor mechanism, then source density should not be credited with explaining these phenomena. Its role must be looked for elsewhere.

In the biological interpretation, the activator u represents a local head-forming signal. A regenerated head is therefore represented, at this level of description, by the emergence of a localized activator peak. The reference model has exactly the phase-space structure needed for a threshold response: the zero state is stable, a lower positive state acts as a threshold, and an upper positive state is the branch from which spatial patterns can form. This makes it possible to distinguish weak perturbations, which decay, from sufficiently strong wound-like perturbations, which trigger regeneration.

If the system is initialized close to the diffusion-unstable homogeneous state, then small spatial perturbations grow and a pattern is formed. This is the standard pattern-formation regime. It shows that the model can create localized activator peaks, but it does not yet represent a regeneration experiment, because the system is already placed near a pattern-forming state.

A regeneration experiment is better represented by starting near the resting state. If the head is removed but no additional wound signal is introduced, the perturbation is too weak to cross the threshold. The solution then returns to the zero steady state, and no new head is formed. This is a useful feature rather than a failure: it means that the model does not regenerate automatically from every small disturbance.

Regeneration appears when the removal experiment is accompanied by a sufficiently strong localized perturbation. Such a perturbation represents a wound signal. If it pushes the solution past the unstable threshold branch, an activator peak can grow and persist. In this sense, the two-equation model already captures the basic logic of wound-induced head formation: injury is not merely a change in geometry, but a local activation event strong enough to move the tissue out of the basin of attraction of the resting state.

The same mechanism can also produce two-head configurations on sufficiently large domains. A second localized peak can be initiated by introducing a wound in the middle of the tissue or near

one of the ends. However, the final position of the new peak is not arbitrary. The model selects preferred patterning positions, for example at a boundary, near the middle, or close to one third of the tissue, depending on the domain size and on the unstable spatial modes available to the system. A wound imposed away from these preferred locations may create a transient peak at the wound site, but this peak can later drift toward a model-selected position.

This behaviour is especially visible when the domain size is varied. On a small domain, simulations often converge to a single boundary peak. If this one-peak solution is continued while the domain is gradually increased, it can remain stable even on much larger domains. By contrast, if a large domain is initialized near the zero state and a strong boundary perturbation is introduced, the system again tends to form a localized boundary peak. But if the same large domain is initialized close to the diffusion-unstable homogeneous state, a small perturbation typically does not select a single head. Instead, the solution converges to a multi-peak pattern.

Thus the observed number and position of peaks are not determined only by the parameters of the equations. They also depend on the domain size, the initial condition, and the path by which the pattern is generated. A one-head pattern obtained by continuation from a small domain is not the same numerical experiment as a pattern selected from small random perturbations of a homogeneous pattern-forming state.

The conclusion is that source density is not needed for these basic regeneration phenomena. The two-equation Gierer–Meinhardt model already accounts for decay of weak perturbations, wound-induced head formation, one- and two-peak patterns, and the selection of preferred peak locations. What it does not provide is an autonomous mechanism for positional memory: the model can create peaks, but it does not remember where a tissue fragment came from, nor does it guarantee that a head remains anchored at an arbitrary graft-induced position. This is the point at which source density becomes relevant.

4. SOURCE DENSITY AND POSITIONAL MEMORY

- Motivation from grafting/polarization experiments:
 - Tissue fragments taken from different body regions can behave differently after being grafted.
 - The outcome depends on how fragments are combined.
 - This suggests positional memory.
- Problem for classical Gierer–Meinhardt models:
 - Away from the head, activator concentration may be very small.
 - Classical Gierer–Meinhardt variables may not encode whether a fragment comes from the upper or lower body column.
 - The model lacks a mechanism to remember tissue origin.
- Source density/source gradient as positional information:
 - ρ can encode distance from the head or positional identity.
 - It is higher near the head and lower away from the head.
 - It may act as a localizer for future head formation.
- First modelling option: prescribed source density $\rho(x)$.
 - $\rho(x)$ is fixed in time.
 - It represents externally imposed positional information.
 - Simulate grafted profiles by combining different pieces of $\rho(x)$.
 - Wounds are introduced near grafting points.
 - Peaks can emerge near maxima or discontinuities/transition regions of ρ .
 - Head positions need not coincide with Turing-selected points.

- Second modelling option: dynamic source density $\rho(t, x)$.
 - Initial condition $\rho(0, x)$ encodes the grafted tissue configuration.
 - ρ evolves slowly.
 - ρ must react slower than activator-inhibitor dynamics to preserve memory.
 - ρ must diffuse:
 - * not too fast, otherwise it becomes nearly constant;
 - * not too slow, otherwise it remains too localized near peaks;
 - * an intermediate diffusion rate can encode approximate distance from heads.
- Expected numerical behavior:
 - initially similar to prescribed $\rho(x)$;
 - peaks form near source-density maxima;
 - peaks can remain there for long times;
 - this creates metastable positional memory;
 - over longer times, peaks may drift toward Turing-selected patterns.
- Main conclusion:
 - Source density helps encode positional memory.
 - However, dynamic source density alone may generate metastable transients rather than genuine stable stationary heads at arbitrary positions.

5. A HEAD IDENTITY EXTENSION

- Problem to solve:
 - Dynamic source density can preserve memory for long times but may not create true stationary anchoring.
 - Activator peaks are transient biochemical signals.
 - A head should represent a more stable morphological commitment.
- Proposed mechanism:
 - Introduce a new non-diffusing variable $h(t, x)$, called head identity.
 - The activator u can induce h , but only if u is high enough for long enough.
 - h then generates or organizes source density ρ .
- Conceptual hierarchy:
 - u : fast activator signal;
 - v : inhibitor;
 - h : slow local head identity/morphological commitment;
 - ρ : positional field generated by heads.
- Key modelling principle:
 - Source density should encode distance from actual heads, not from temporary activator peaks.
- Hysteresis:
 - head formation threshold is higher than head maintenance threshold;
 - it is harder to create a head than to maintain an existing head;
 - after h is established, it does not disappear immediately when u decreases.
- Possible implementation:
 - h does not diffuse;

- h grows when u exceeds a high threshold;
- h decays when u falls below a lower threshold;
- use a smooth bistable or hysteretic response function.
- Expected behavior:
 - wound-induced activator peaks can create a head if strong and persistent enough;
 - once a head is established, it anchors source density;
 - peaks can remain at graft-induced positions rather than only Turing-selected positions;
 - heads too close to each other may still compete and one may disappear.
- Main claim:
 - Separating activator peaks from head identity provides an autonomous mechanism for positional memory.

6. DISCUSSION AND CONCLUSION

- The model audit separates basic pattern generation from positional memory. Suitable two-variable Gierer–Meinhardt-type systems already provide a stable resting state, threshold activation, and localized activator peaks.
- Source density should therefore not be treated as a generic ingredient that explains regeneration by itself. Its main role is to encode positional information, especially in grafting and polarization experiments.
- Prescribed source density acts as an external positional cue, while dynamic source density can provide slow memory. However, dynamic source density alone may produce metastable anchoring rather than genuinely stable heads at arbitrary graft-induced positions.
- Introducing a head identity variable with hysteresis separates transient activator peaks from stable morphological commitment. In this interpretation, activator signals are fast, head identity records downstream commitment, and source density is organized by established heads.
- Future work should complete the stability and bifurcation analysis, add numerical grafting experiments, calibrate the model biologically, and compare the proposed mechanism with experimental data.

REFERENCES